

BUDDY: An Intelligent Robot Dog for Offline Voice Interaction, Visual Perception, and Safety-Aware Multimodal Control

COMMON INTEGRATED SOLUTION REPORT

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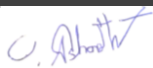
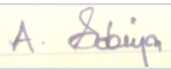
Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

BUDDY is an affordable intelligent robot-dog prototype investigating how local human-robot interaction, visual perception, environmental sensing, autonomous-behaviour research, and physical actuation can be integrated on Raspberry Pi-class hardware while preserving an explicit safety boundary around movement. The solution combines three research components: offline voice intelligence and safety-aware system integration; visual intelligence for person recognition, facial-expression cues, unknown-person handling, adaptive object learning, and fall-state reasoning; and environment-aware navigation, obstacle avoidance, human following, and 2D path-planning research. ROS 2 Jazzy provides the shared middleware, while an independent safety supervisor validates movement requests against current ultrasonic observations before approved commands reach the motor executor. A Flutter/Flask application exposes visual status, alerts, enrollment, and camera media to the user. Evaluation combined physical integration tests, subsystem demonstrations, software verification, and simulation. The physical prototype demonstrated automatic startup, offline wake-word interaction, spoken forward, backward, left, right, and stop commands, four-direction ultrasonic sensing, dynamic obstacle-mediated stopping and resumption, and camera-based face following. The visual subsystem additionally demonstrated known/unknown-person handling, expression cues, unknown-person alert and enrollment, adaptive learning of a mug, box, and phone, and camera streaming/recording. A* planning, field-of-view reasoning, follow-me behaviour, and dynamic replanning were demonstrated in simulation. Overall, the results support the feasibility of a modular low-cost companion robot while identifying remaining work in autonomous patrol, fall validation, and fully concurrent multimodal operation.

Keywords: companion robotics, multimodal interaction, ROS 2, edge AI, safety-aware control

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
ASR	Automatic Speech Recognition
CNN	Convolutional Neural Network
FER	Facial Expression Recognition
FOV	Field of View
HRI	Human-Robot Interaction
LBPH	Local Binary Patterns Histograms
ML	Machine Learning
ROS2	Robot Operating System 2
STT	Speech-to-Text
TTS	Text-to-Speech

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Companion robots combine physical embodiment with sensing, interaction, autonomy, and actuation. Unlike a software-only assistant, an embodied companion must interpret human requests while continuously accounting for the physical environment in which movement occurs. This makes system integration a central research concern: speech recognition, visual perception, navigation, and application interfaces are useful only when their outputs can be coordinated without creating unsafe or unpredictable physical behaviour [1]-[4].

BUDDY was developed as an affordable intelligent robot-dog prototype using a Raspberry Pi 4, ROS 2 Jazzy, four-wheel mobility, four-direction ultrasonic sensing, a two-axis camera head, local speech processing, visual perception, and a companion application. The project is intentionally modular. Each research component can be developed independently, but the final value arises from the complete system: the robot can listen, observe, reason about environmental conditions, communicate status, and perform safety-supervised physical actions.

The project also addresses practical constraints of student-scale and household-oriented robotics. Continuous cloud dependence can reduce privacy and availability, expensive perception hardware can make prototypes inaccessible, and tightly coupled code can make integration fragile. BUDDY therefore emphasises local processing where practical, commodity components, explicit interfaces, persistent but reviewable local data, and a separation between high-level intelligence and safety-critical motor authority.

1.2 Stakeholders and user needs

The primary stakeholders are household users who interact with the robot, project developers and researchers who need a reproducible modular platform, and academic evaluators concerned with research validity, safety, privacy, and technical contribution. Secondary stakeholders include educational robotics users and future

developers who may extend the platform with improved localisation, perception, or interaction capabilities.

Stakeholder	Primary need	How BUDDY addresses it
Household user	Natural interaction, understandable status, safe movement, local privacy	Wake-word voice interaction, visual recognition/alerts, app status, safety-supervised motion
Developer/researcher	Modular architecture and testable interfaces	ROS 2 packages, 14 custom interfaces, hardware abstraction, separate visual API
Academic evaluator	Clear evidence and distinction between completed and pending work	Component-level and integrated validation, explicit limitations, no fabricated metrics
Future adopter/educator	Affordable and extensible platform	Raspberry Pi-class compute, commodity sensors, open software stack

Table 1.1 Stakeholders and confirmed project needs

1.3 Alignment with the Sustainable Development Goals

BUDDY has plausible alignment with Sustainable Development Goal 9 (Industry, Innovation and Infrastructure) through its exploration of affordable edge-AI robotics and modular embedded-system engineering. It also has potential relevance to Sustainable Development Goal 3 (Good Health and Well-being) through companionship-oriented interaction and safety-support features such as fall-state reasoning. However, the current research does not claim measured health, wellbeing, or societal impact. These are intended application directions rather than validated outcomes.

1.4 Purpose and scope of the common report

This common report presents the complete BUDDY solution, the relationships among the three individual research components, the shared interfaces and infrastructure, the integrated testing evidence, and the combined value of the prototype. Detailed algorithms, component-specific experiments, and individual contribution evidence are presented in the separate theses of Ashandth Uthayashankar, Sobiya Anton Suresh, and S. S. Y. Wickramasinghe. The common report therefore focuses on the integrated problem and system rather than repeating three individual dissertations.

2 OVERALL PROBLEM DEFINITION AND THREE SUBPROBLEMS

2.1 Overall problem definition

Affordable companion robots must combine natural human interaction, environmental awareness, visual context, autonomous behaviour, and physical actuation on constrained hardware. The common problem is not the absence of individual algorithms for speech, face recognition, or path planning; mature techniques already exist for each. The difficult problem is integrating these capabilities into one embodied platform while preserving responsiveness, understandable behaviour, privacy, and a safe boundary between uncertain high-level decisions and physical motor execution.

In particular, a recognised speech command or autonomous navigation intention should not automatically be equivalent to motor authority. The physical environment can change after a command is issued, sensor information can become stale, and high-level classifiers may make incorrect decisions. BUDDY therefore treats movement as a request that must be independently validated against current environmental state before it can reach the motor drivers.

2.2 Root causes and overall research gap

The project identifies five interacting causes of the integration problem. First, multimodal perception consumes limited CPU, memory, and I/O resources on Raspberry Pi-class hardware. Second, sensing is uncertain: microphones are affected by noise and distance, cameras by lighting and pose, and low-cost ultrasonic sensors by geometry and cross-interference. Third, the individual components operate at different time scales and produce different data types. Fourth, physical actuation introduces safety consequences that do not exist in a software-only assistant. Fifth, users need a simple way to understand robot state, review security events, and intervene when automatic behaviour is inappropriate.

The resulting research gap is a systems-integration gap: how to combine local speech, visual context, navigation research, environmental safety sensing, user-facing monitoring, and physical movement in an affordable companion robot without

allowing uncertain high-level intelligence to directly control actuators. BUDDY addresses this gap through three component streams coordinated by shared ROS 2 interfaces and an independent safety-supervision layer.

2.3 Subproblem and component mapping

Subproblem	Component owner	Core methods	Integrated output
Natural local interaction and safe translation of speech into robot actions	Ashandth Uthayashankar	Vosk wake/STT, deterministic motion grammar, ML intent fallback, Piper TTS, persistent memory, ROS 2 integration	Movement requests, spoken responses, remembered user context
Visual awareness of people, expressions, unfamiliar visitors, objects, and safety-related body state	Sobiya Anton Suresh	OpenCV, hybrid identity recognition, DeepFace, MobileNetV2 features, MediaPipe Pose, Flask + Flutter	Recognised/unknown person state, expression cue, object memory, alerts, camera/media services
Environment-aware movement, obstacle handling, human following, and route planning	S. S. Y. Wickramasinghe	Ultrasonic sensing, A* grid planning, PID following, FOV/occlusion reasoning, state-machine recovery	Navigation intentions, obstacle-aware behaviour, follow/patrol research path

Table 2.1 Mapping of the three subproblems to individual components

2.4 Overall aim and group objectives

The overall aim is to design, implement, and evaluate an affordable multimodal companion robot that can interact locally with users, perceive visual and environmental context, and perform physical movement through a modular safety-supervised architecture.

Objective	Mapped component(s)	Evidence used
O1. Provide practical offline voice interaction and deterministic voice-controlled movement	Voice + shared safety	Cold-boot test, wake-word range observations, physical movement commands
O2. Provide person-aware visual interaction and user-reviewable visual events	Visual + application	Known/unknown trials, emotion output, alert/enrollment screens, camera/media tests
O3. Provide obstacle-aware movement and a foundation for autonomous indoor behaviour	Navigation + shared safety	Four ultrasonic channels, dynamic obstacle stop/resume, A* and follow-me simulation
O4. Separate high-level intent from physical motor authority	All / shared architecture	/movement/request -> safety supervisor -> /movement/safe -> motor executor
O5. Integrate the subsystems through reusable interfaces and deployment tooling	All	ROS 2 packages/interfaces, systemd startup, hardware abstraction, API integration

Table 2.2 Common objectives and evaluation evidence

2.5 Scope assumptions and constraints

The present prototype is intended primarily for controlled indoor environments. It uses a wheeled base rather than legged locomotion, does not claim safety certification, and does not yet provide complete autonomous localisation and mapping. The visual subsystem was tested at small project scale rather than as a population-level biometric system. Fall detection is implemented but awaits credible full-body physical validation. The app is operational with Sobiya's subsystem backend, while final integration of every app function with the completed robot stack remains continuing work.

The voice path is validated using a single AB13X USB microphone. Dual-identical-microphone localisation remains experimental because reliable simultaneous acquisition was not established on the Raspberry Pi. Navigation simulation demonstrates route planning and follow behaviour, but complete patrol/localisation is not reported as physically validated. These boundaries are maintained throughout the report to avoid converting software implementation or simulation into unsupported end-to-end claims.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

BUDDY is organised as a layered multimodal system. Audio, camera, and ultrasonic sensors provide raw observations. Component-specific software interprets those observations and produces high-level state or movement requests. Shared ROS 2 interfaces transport robot state and commands. The safety supervisor acts as the gatekeeper for physical movement, while a motor executor and hardware abstraction layer control the four-wheel platform. A Flutter application exposes visual status, camera media, alerts, household enrollment, and selected controls to the user.

The architectural principle is that perception does not directly own actuation. Voice may request movement, navigation may generate an intended motion, and visual processing may request head orientation or generate interaction context, but physical wheel movement passes through a safety boundary. This enables the robot to reevaluate the environment continuously instead of assuming that a command remains safe simply because it was valid when first issued.

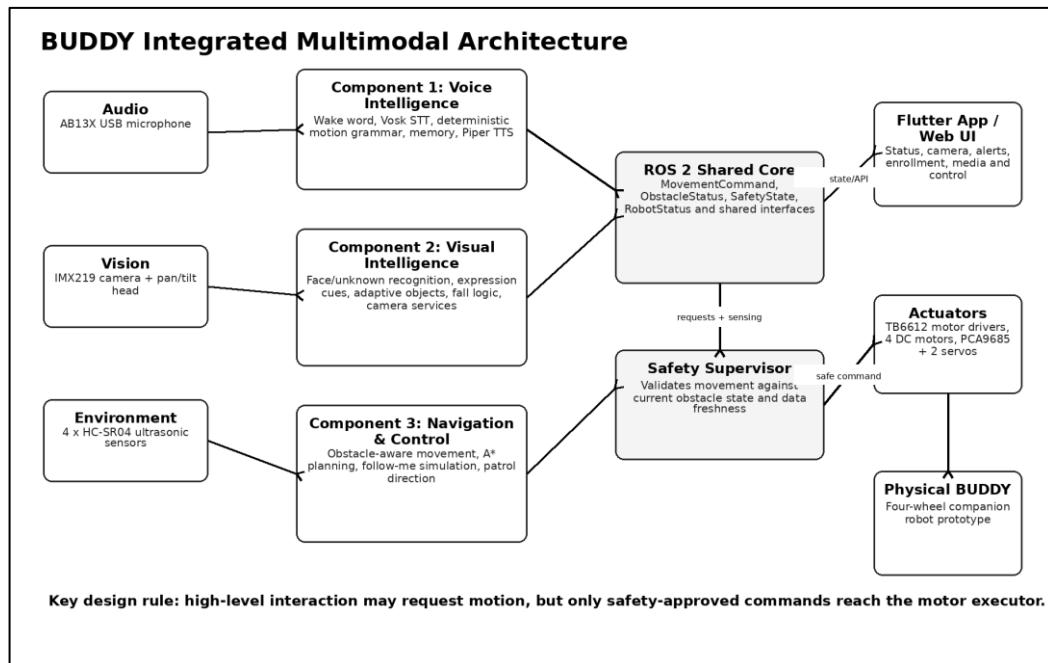


Figure 3.1 High-level architecture of the integrated BUDDY solution

3.2 High level architecture

The shared software foundation is built around ROS 2 Jazzy. The repository contains the `buddy_core` package for common robot behaviour and hardware services, `buddy_interfaces` for formal message/service/action contracts, and `buddy_bringup` for startup structure. The interface package defines fourteen custom contracts: eight messages, three services, and three actions. These include `MovementCommand`, `ObstacleStatus`, `SafetyState`, `RobotStatus`, `VoiceCommand`, `RecognizedPerson`, `EmotionObservation`, `SecurityEvent`, `EmergencyStop`, `ResetEmergencyStop`, `SetRobotMode`, `FollowPerson`, `ApproachCaller`, and `PatrolRoute`.

The application-facing visual subsystem additionally uses a Flask HTTP API and MJPEG stream. This API boundary allows the Flutter client to obtain notifications, media, identity/emotion status, household enrollment, and camera functions without coupling the mobile UI directly to OpenCV or robot internals. The coexistence of ROS 2 and HTTP is deliberate: ROS 2 is used for robot-internal coordination, while HTTP provides a convenient user-interface boundary.

3.3 Component 1 summary - Offline voice intelligence and system integration

Owner: Ashandth Uthayashankar (IT22272522). The subproblem is enabling natural, local speech interaction while ensuring that spoken or conversational output cannot directly control the motors. Audio is captured through an AB13X USB microphone at 48 kHz stereo, averaged and converted to 16 kHz mono for Vosk. A dedicated wake recogniser activates on “Buddy”, while a constrained command grammar handles safety-relevant motion requests. General conversation is handled separately through deterministic responses, persistent memory, and an ML intent fallback. Piper provides local TTS. The component publishes `MovementCommand` requests rather than writing directly to GPIO.

Ashandth also led significant shared integration work around ROS 2, safety supervision, motors, ultrasonic sensing, deployment, power/hardware troubleshooting, and automatic startup. These shared engineering tasks are reported as integration contributions rather than as ownership of the other members' research components.

3.4 Component 2 summary - Visual intelligence and companion application

Owner: Sobiya Anton Suresh (IT22203694). The visual component expands beyond basic face recognition into person identity, unknown-person handling, facial-expression cues, pan/tilt face tracking, adaptive object learning, pose-based fall reasoning, camera media services, and a Flutter companion application. The implementation combines OpenCV face detection, local cosine/LBPH identity memory, DeepFace emotion analysis, MobileNetV2 feature extraction for user-taught objects, and MediaPipe Pose for body-landmark reasoning.

At subsystem level, the visual stack demonstrated known and unknown user handling, live expression labels, unknown-person security notifications and explicit enrollment, adaptive learning of a mug, box, and phone, and camera streaming/recording. Known users can occasionally be rejected as UNKNOWN under unfavourable image conditions; this is recorded as a limitation. The fall module is implemented, but final full-body physical validation remains pending.

3.5 Component 3 summary - Navigation, obstacle avoidance, human following and path planning

Owner: S. S. Y. Wickramasinghe (IT21816086). The navigation component addresses movement through indoor environments, obstacle-aware behaviour, human-following logic, patrol direction, and route planning. His navigation implementation in the R26-SE-040 repository contains an Ursina-based 3D-visualised 2D navigation simulation with room geometry, furniture obstacles, a cost-aware occupancy grid, A* search, line-of-sight checks, field-of-view reasoning, PID-style distance control, human-following heel positions, search/recovery states, and dynamic replanning.

Physical evidence available at group level validates the four directional ultrasonic channels and obstacle-mediated stopping/resumption. The broader A*, follow-me, and patrol logic is reported as simulation/development evidence rather than as complete physical autonomy. Future mapping work can use ultrasonic range measurements together with monocular camera cues to maintain an occupancy-style 2D representation, but the Raspberry Pi Camera Module 2 is not treated as a true depth sensor or 3D point-cloud source.

3.6 Shared application and integration layer

The companion application is primarily implemented by Sobiya, with navigation/security requirements and project-website contribution from Yohan, and final robot-stack integration supported by Ashandth. It provides a user-facing view of robot status, live camera, media, identity/emotion information, unknown-person events, enrollment, activity, settings, and control concepts. The application is not treated as a fourth independent research component; it is a shared presentation and interaction layer through which the three components become understandable and controllable to the user.

3.7 End to end user and data flow

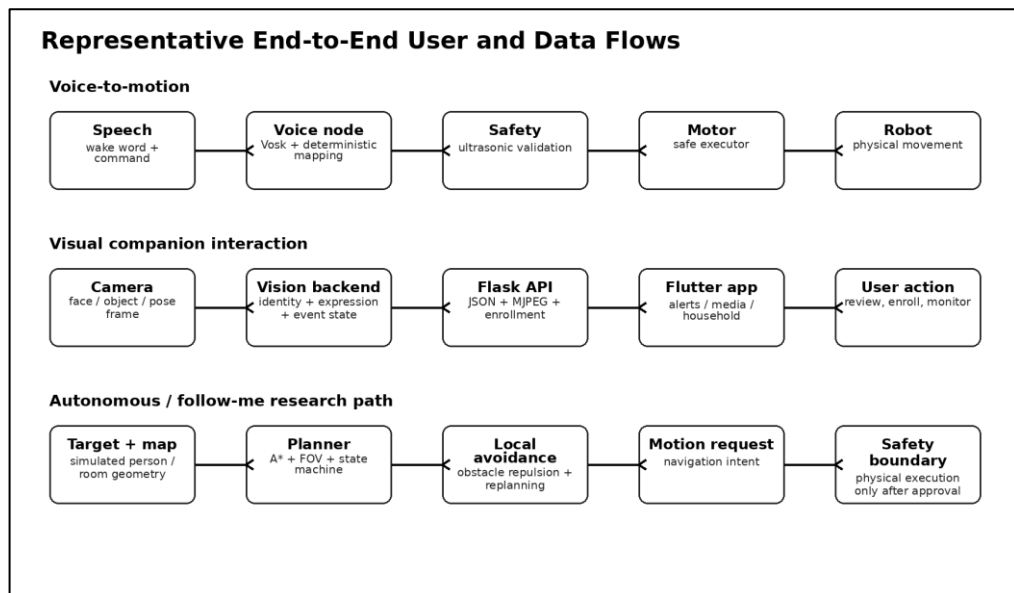


Figure 3.2 Representative end-to-end user and data flows across the three components

Three representative flows demonstrate the integration. In the voice-to-motion flow, speech is recognised and converted into a structured movement request; current obstacle state is then used to approve or deny physical execution. In the visual-interaction flow, camera frames become identity, expression, object, or unknown-person state; Flask exposes that state to the Flutter application, where a user can review an event or enroll a person. In the autonomous/follow-me research flow, target position and map information are transformed into path-planning and local-avoidance intentions before any eventual physical command reaches the safety boundary.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

Integration followed a staged approach: establish stable hardware interfaces; validate motors and sensors independently; formalise shared ROS 2 contracts; implement a safety supervisor between request sources and the motor executor; validate voice-to-motion and obstacle intervention physically; integrate pan/tilt visual tracking; and then connect richer visual/application functions and navigation research. This reduced the risk of debugging perception, networking, actuation, and power simultaneously.

The team also separated validated runtime code from experimental extensions. Hardware abstraction defaults to mock/safe behaviour unless physical hardware is explicitly enabled. The voice and visual components can be exercised independently on development machines. Navigation logic can be evaluated in simulation before moving to the physical platform. This modularity supports regression testing and limits the effect of one subsystem failure on the rest of the robot.

4.2 Interfaces, data contracts and shared models

Interface / boundary	Producer	Consumer	Purpose / rule
/movement/request (MovementCommand)	Voice, navigation, app/manual sources	Safety supervisor	Represents intent only; no direct motor authority
ObstacleStatus	Ultrasonic sensor node	Safety supervisor / navigation	Four-direction environment state plus freshness
/movement/safe (MovementCommand)	Safety supervisor	Motor executor	Only approved movement reaches physical motor execution
SafetyState	Safety supervisor	Shared robot/application state	NORMAL/CAUTION/RESTRICTED/EMERGENCY status
Head/neck control request	Visual tracking	PCA9685 neck controller	Pan/tilt orientation within verified servo limits
Flask JSON/MJPEG API	Visual backend	Flutter client	Identity, emotion, alerts, enrollment, live media and status
Persistent JSON stores	Visual/voice modules	Same subsystem after restart	Known faces, object features, emotion/user memory

Table 4.1 Principal integration interfaces and data contracts

4.3 Shared nonfunctional requirements

Requirement	System response
Safety	Movement requests are independently validated; stale obstacle state fails toward STOP.
Privacy	Core voice and visual functions favour local processing and local data stores where practical.
Responsiveness	Frame skipping/caching, background tasks, lightweight models, and constrained voice grammar reduce latency.
Reliability	Watchdogs, timeouts, exception handling, fallback recognition paths, and auto-restart support operation.
Maintainability	ROS 2 packages, hardware abstraction, frozen interfaces, configuration files, and version control isolate responsibilities.
Usability	Wake-word interaction and companion-app status/alerts provide understandable user control.
Transparency	Incomplete features are labelled as software-level, simulated, or pending rather than reported as validated.

Table 4.2 Shared nonfunctional requirements

4.4 Integrated workflow and exception handling

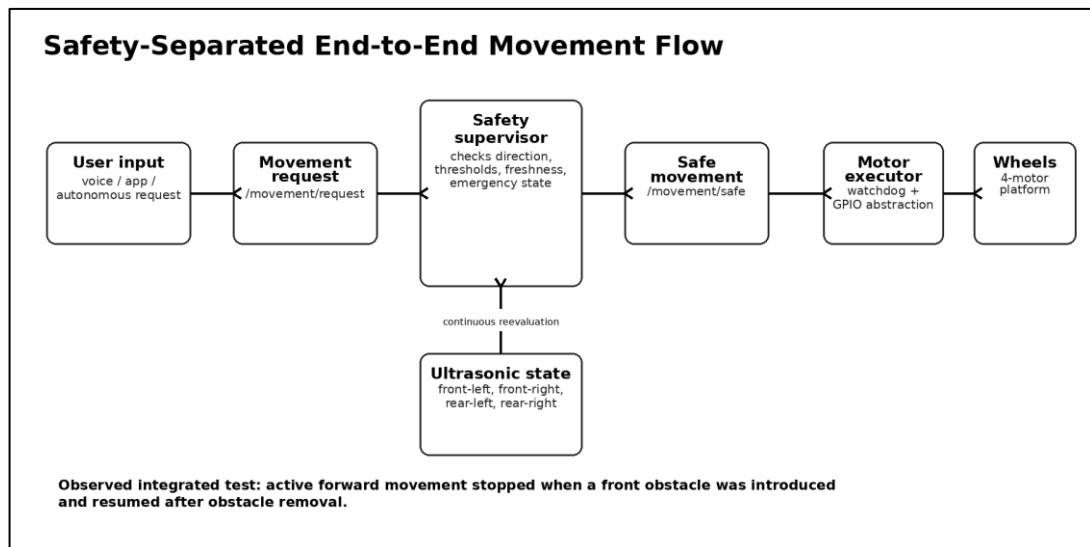


Figure 4.1 Safety-separated end-to-end movement workflow

Normal movement begins with a high-level source producing a MovementCommand request. The safety supervisor evaluates the requested direction against recent ultrasonic observations and current safety state. Approved movement is republished on the safe topic, which is the only movement source consumed by the motor

executor. During active movement, the request is reevaluated when environmental state changes. Obstacle information older than approximately one second is treated as stale; the safety layer therefore stops rather than assuming that an outdated clear reading remains valid.

The motor executor adds a separate short watchdog, providing another defensive boundary if command updates stop unexpectedly. Similarly, the visual subsystem uses repeated unknown observations and a cooldown before generating an alert, reducing transient false alarms. The voice subsystem separates the wake recogniser from the active command/general recognisers and clears queued audio after speech output to reduce self-triggering. These mechanisms show a common project pattern: uncertainty is managed through explicit state, persistence checks, freshness rules, and safe fallbacks.

4.5 Team contribution and dependency matrix

Area	Ashandth Uthayashankar	Sobiya Anton Suresh	S. S. Y. Wickramasinghe
Primary research	Offline voice intelligence; safety-aware integration	Visual intelligence; app functionality	Navigation; obstacle avoidance; human following; path planning
Core implementation	Vosk/Piper pipeline, ML intent, memory, ROS 2 voice, safety/motor/sensor integration	Face/emotion/object/fall modules, Flask services, Flutter app	A* planner, 3D-visualised 2D follow-me simulation, state machine, obstacle/navigation logic
Physical integration	Motor, sensor, power, startup and cross-component troubleshooting	Camera-head/vision integration with team support	Navigation sensing and movement direction integration
Shared outputs	Research paper, architecture, final robot integration	Research paper, app/screens, vision evidence	Research paper, website/UI direction, navigation evidence
Dependency on others	Uses visual/navigation state through shared system	Uses robot speech/head/application integration	Uses shared safety and motor execution

Table 4.3 Team contribution and dependency matrix

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

Layer	Technologies / hardware	Integrated role
Embedded compute	Raspberry Pi 4B 4 GB, Ubuntu 24.04 ARM64	Runs ROS 2 and core robot services
Robot middleware	ROS 2 Jazzy, custom messages/services/actions	Inter-component state, requests, safety, actions
Voice	Python, sounddevice, Vosk, Piper, scikit-learn	Offline wake/STT, deterministic movement grammar, TTS, intent fallback
Vision	Python, OpenCV, DeepFace, MobileNetV2, MediaPipe	Identity, expression, object memory, pose reasoning, camera services
Navigation	Python, Ursina simulation, A*, PID-style following	Route planning, follow-me and state-machine research
Application	Flutter/Dart + Flask/CORS + MJPEG	User-facing status, alerts, enrollment, camera/media and control
Actuation	2 x TB6612FNG, 4 geared motors, PCA9685, 2 x MG90-class servos	Wheeled movement and pan/tilt head
Environment sensing	4 x HC-SR04 via level shifting	Front/rear left/right obstacle state
Deployment	systemd service, shell startup script, YAML configuration	Automatic startup and reproducible runtime configuration

Table 5.1 Common technology stack

5.2 Development and integration process

Development used a shared GitHub repository with feature branches and an integration branch. Work was divided by research ownership but converged through stable shared interfaces. The ROS 2 workspace separates core robot functionality from interface definitions and bringup structure. Configuration files hold physical/simulation parameters so that the same codebase can be exercised with mock or real hardware. This reduced the need for direct code edits when moving between laptop development, Raspberry Pi testing, and physical motor/sensor operation.

Component work was integrated incrementally. The team first validated low-level hardware assumptions, then movement and ultrasonic sensing, then the independent safety layer, then voice-to-motion, pan/tilt visual tracking, and richer perception/application functions. Yohan's navigation branch additionally provided a simulation environment for route and follow behaviours that could be developed without risking collisions on the physical robot.

5.3 Deployment and operational configuration

The Raspberry Pi runtime is configured to start automatically through a systemd service. The startup script waits for the operating system, sources ROS 2 and the workspace environment, activates the required Python environment, and then launches the safety supervisor, physical ultrasonic node, motor executor, and voice process in sequence. A cold-boot experiment verified that the system reached an interactive state without manual SSH or terminal launch and produced the spoken readiness message before responding to the wake phrase.

The physical power architecture separates Raspberry Pi computing power from the higher-current motor/servo rail. Three protected 18650 cells, battery-management/protection, fuse, switching, and DC-DC regulation supply the actuation rail, while common ground is maintained. Servo and motor loads are therefore not powered through Raspberry Pi GPIO. This separation is important for embedded reliability because motor and servo current transients can otherwise destabilise compute and sensor operation.

5.4 Evidence of team execution

Execution evidence is distributed across Git history, the research paper, the PP2 presentation, individual theses, component screenshots, source code, and physical test records. The voice/integration branch contains the offline speech pipeline and ROS 2 deployment files. The visual branches and Flutter repository contain recognition, object learning, fall logic, unknown-person workflows, camera/media services, and the application. Yohan's navigation implementation in the R26-SE-040 repository contains the 3D-visualised 2D follow-me and A* navigation simulation, while a separate project website documents the BUDDY research direction and team roles.

The team also maintained a deliberate evidence rule: software implementation, simulation, and physical validation are not treated as equivalent. This distinction is visible in the final reporting of dual-microphone localisation, fall detection, full autonomous patrol, and final app-to-complete-robot integration, all of which remain partial or continuing rather than being converted into unsupported success claims.

6 INTEGRATED TESTING, EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

Group-level evaluation focuses on whether the principal multimodal pathways function coherently and whether the safety architecture behaves as intended. The strongest success criteria are observable end-to-end behaviours: automatic startup; wake and spoken movement; four-direction environmental sensing; dynamic safety intervention; active camera-head tracking; user-visible visual events; and consistent separation between development/simulation evidence and physically validated evidence.

Evaluation question	Success criterion	Evidence source
Can BUDDY start as an embedded system?	Robot runtime starts after cold boot without manual terminal launch	systemd cold-boot test
Can local speech control physical movement?	Wake + forward/backward/left/right/stop produce expected wheel behaviour	physical voice-motion trials
Can safety override an active request?	Forward motion stops when obstacle appears and can resume after removal	dynamic obstacle experiment
Can the robot perceive users visually?	Face following and known/unknown/expression outputs are observable	physical head test + visual subsystem trials
Can users review visual events?	Unknown event appears with captured face and explicit enrollment action	Flutter/Flask app evidence
Can autonomous behaviour be developed safely before deployment?	A* and follow-me logic operate in simulation with obstacles and state recovery	Ursina navigation simulation

Table 6.1 Integrated evaluation questions and success criteria

6.2 Integration and end to end testing

The most important end-to-end physical test exercised the complete safety-separated movement path. BUDDY received a continuous forward request and began moving. A physical obstacle was then introduced into the front ultrasonic sensing region. The updated environmental state caused the still-active request to be reevaluated, a STOP command reached the motor executor, and forward movement ceased. When the obstacle was removed, the still-active request could be reconsidered and movement resumed. This demonstrates continuous environmental reevaluation rather than a one-time safety check at command initiation [1].

The voice path was also exercised from cold boot. The local wake phrase “Buddy” remained usable within the tested 0-3 m range and became inconsistent at approximately 3.4 m. Forward, backward, left, right, and stop commands produced the expected physical wheel behaviour. A controlled multi-participant speech-recognition study was not completed, so the group does not report a generalized recognition-accuracy percentage.

The visual perception-to-actuation path was evaluated independently by presenting a face at different positions in the camera frame. Pan and tilt corrections caused the physical camera head to follow the face. Separately, Sobiya's visual subsystem demonstrated known/unknown identity, expression output, unknown-person notifications and enrollment, object learning, and camera media functions through the application. These tests establish component and subsystem capability but do not imply that every visual function was active simultaneously with the full autonomous robot runtime.

6.3 Overall results

Result area	Observed evidence	Validation status
Automatic embedded startup	Runtime and voice start after cold boot without SSH	Physically validated
Wake-word interaction	Usable at 0-3 m; inconsistent around 3.4 m in tested environment	Physically observed
Voice-controlled movement	Forward, backward, left, right, stop	Physically validated
Four-direction ultrasonic sensing	Front-left, front-right, rear-left, rear-right channels available	Physically validated
Dynamic safety intervention	Active forward movement stopped when obstacle introduced; resumed after removal	Physically validated
Face-following head	Camera head followed changes in face position	Physically validated
Known/unknown person handling	~5 known people; ~20 unfamiliar image/video samples plus live cases	Subsystem functional evidence
Expression cues	Seven-category live output available	Subsystem functional evidence
Unknown alert and enrollment	Captured face shown in app with registration action	Subsystem functional evidence
Adaptive object learning	Mug, box, phone learned/tested	Subsystem functional evidence
Camera stream/recording	Live stream, recording and media retrieval demonstrated	Subsystem functional evidence
Fall detection	Pose-based module implemented; full-body physical validation pending	Implemented / pending validation
A* / follow-me / recovery	3D-visualised 2D simulation with grid planning, FOV, PID following, dynamic replanning	Simulation validated

Complete patrol/localisation	Not yet demonstrated physically	Continuing work
Software tests	13 relevant movement/safety tests passed in project paper	Software-level validated

Table 6.2 Synthesised group-level results

6.4 Stakeholder and user validation

User validation at this stage is primarily functional and project-scale rather than a formal usability study. Visual recognition was exercised with approximately five known individuals and around twenty unfamiliar image/video samples. Unknown-person events were reviewed through the application, demonstrating that the user can make an explicit trust decision before a face becomes enrolled. Object learning was exercised with three everyday objects. These observations are sufficient to demonstrate intended workflows but are not large enough to support population-level biometric, affect, or usability claims.

The principal stakeholder feedback shaping the project came from academic supervision. One important example was the decision to broaden the visual component beyond basic face recognition into adaptive object learning, unknown-person handling, fall logic, and richer application integration. At group level, the project similarly evolved from isolated component prototypes toward explicit safety separation and stronger evidence discipline.

6.5 Achievement of common objectives

Objective	Achievement	Justification
O1 Offline voice interaction and deterministic movement	Achieved at prototype level	Wake, local STT/TTS and five physical movement commands demonstrated.
O2 Person-aware visual interaction and user-reviewable events	Achieved at subsystem level	Known/unknown, expression, alerts, enrollment, camera/media and object learning demonstrated.
O3 Obstacle-aware movement and foundation for autonomy	Partly achieved	Physical obstacle safety demonstrated; A*/follow-me simulated; full patrol/localisation pending.
O4 Separate high-level intent from motor authority	Achieved	Safety supervisor and safe movement topic provide explicit independent boundary.
O5 Integrate components through reusable interfaces and deployment tooling	Partly to substantially achieved	ROS 2 interfaces, systemd startup, app API and shared hardware validated; final all-feature concurrent integration remains continuing.

Table 6.3 Achievement of common objectives

6.6 Integrated limitations and threats

The main integrated limitations are resource contention on Raspberry Pi-class hardware, the absence of wheel encoders/IMU for precise closed-loop localisation, low-cost ultrasonic sensing limits, camera sensitivity to lighting and pose, small visual test samples, and incomplete full-system concurrency testing. Known users can occasionally be false-rejected as UNKNOWN. The fall detector still requires a suitable full-body camera test. App networking currently depends on development/local-network configuration. Dual-identical-USB-microphone acoustic localisation was not physically validated. The navigation simulation has direct access to known obstacle geometry and simulated target coordinates, whereas the physical robot does not yet possess equivalent global map knowledge or metric localisation; therefore, simulation-level A* and follow-me results should not be interpreted as equivalent physical autonomy.

Validity threats include opportunistic rather than controlled participants, limited repeated trials, use of simulation for advanced navigation, and the possibility that subsystem tests do not capture interactions that appear when all AI workloads execute simultaneously. The group therefore reports functional prototype feasibility rather than generalized reliability, safety certification, clinical benefit, or production readiness.

7 COMMERCIALISATION, SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

BUDDY's value proposition is a modular companion-robot platform that combines local voice interaction, person-aware visual features, safety-supervised movement, and extensible autonomous behaviour using accessible components. Potential beneficiaries include educational robotics programmes, university research laboratories, makers, and households interested in an experimental companion/security platform. The present prototype is not a medical device or certified consumer product.

7.2 Market alternatives and differentiation

The project differentiates itself primarily through integration and openness rather than through a claim of superior performance to commercial robots. Expensive commercial companion platforms may offer polished mechanics and proprietary perception stacks, while simple educational kits are affordable but often lack integrated multimodal intelligence. BUDDY explores the middle ground: an affordable research platform with offline voice, user-reviewable visual awareness, active environmental safety, a user-facing application, and a clear separation between high-level intent and physical authority.

7.3 Business model and feasibility

A realistic sustainability model would initially focus on research/education kits, open modular software, paid assembly/support, or institution-oriented workshops rather than immediate mass consumer deployment. The architecture already separates commodity hardware, software modules, and user-facing services, which supports staged upgrades. However, a final bill of materials, assembly time study, reliability testing, certification costs, and support requirements would be necessary before any production pricing or profitability claim could be justified.

7.4 Adoption, intellectual property and compliance

Adoption barriers include reliability expectations for a physical robot, data privacy around face images and household identity, network security for remote camera/control functions, mechanical durability, battery safety, and the need for transparent limitations around emotion interpretation and fall detection. A production version would require explicit consent and retention controls, encrypted communications, authentication/authorization, software-update procedures, and appropriate safety/risk assessment. Personal-care robot safety standards such as ISO 13482 provide relevant professional context, but BUDDY is not claimed to be certified [13].

The project combines original integration code with open-source libraries and frameworks. Any commercialisation would therefore require a complete dependency/license audit and clear ownership agreements for team code, models, media, and application assets. The privacy-sensitive local databases for known faces, emotion history, and user profile would also require formal data-protection policies.

7.5 Sustainable Development Goal contribution

The strongest defensible SDG connection remains SDG 9 through accessible innovation in embedded robotics and modular local AI. SDG 3 is a potential application connection through companionship and safety-support features, but the project has not measured health or wellbeing outcomes. Future work should evaluate benefits and possible negative effects, including over-reliance on automated emotional interpretation, privacy risks, electronic waste, battery lifecycle, and accessibility for users with different abilities.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

BUDDY demonstrates that an affordable Raspberry Pi-class companion robot can combine offline auditory interaction, visual awareness, environmental sensing, user-facing monitoring, and physical mobility within a modular architecture that preserves a meaningful safety boundary around movement. The strongest integrated result is not any single classifier or planner; it is the coordinated system behaviour in which high-level movement intent is treated as a request and an independent safety layer can interrupt an already-active command when the physical environment changes.

The physical prototype validated automatic startup, local wake-word interaction, spoken movement, four-direction ultrasonic sensing, dynamic obstacle-mediated stopping and resumption, and camera-based face following. The visual subsystem further demonstrated identity/unknown handling, expression cues, alerts and enrollment, adaptive object learning, and camera media functions. Navigation research demonstrated A* path planning, field-of-view reasoning, follow-me states, local avoidance, and dynamic replanning in simulation. Together, these results establish functional feasibility while clearly identifying the remaining work required for complete autonomous patrol and fully concurrent multimodal operation.

8.2 Principal integrated contribution

The principal integrated contribution is a low-cost multimodal robot architecture in which interaction, perception, planning, and actuation are deliberately separated through explicit interfaces and safety supervision. This architecture provides a practical foundation for extending BUDDY without giving conversational AI, facial analysis, or autonomous planning unrestricted motor authority. A second contribution is the project's evidence discipline: software implementation, simulation, subsystem demonstration, and physical validation are recorded as different evidence levels, improving the credibility and reproducibility of the final system report.

8.3 Future directions

- Complete physical validation of the fall-detection module using a full-body camera setup and controlled fall/non-fall trials.
- Integrate every Flutter application function with the final Raspberry Pi runtime using configurable service discovery and authenticated/encrypted communication.
- Add wheel encoders and/or an IMU, and evaluate localisation, route execution, and autonomous patrol using quantitative completion/collision metrics.
- Investigate more capable depth/ranging hardware or robust sensor fusion for 2D mapping; treat monocular Camera Module 2 imagery as visual cues rather than direct depth.
- Conduct controlled multi-speaker voice trials across noise levels and accents, and measure end-to-end latency and resource usage.
- Use improved multi-channel audio hardware if acoustic source localisation is retained as a research extension.
- Run long-duration integrated tests covering power consumption, thermal behaviour, software recovery, network interruption, and concurrent voice/vision/navigation workloads.
- Strengthen privacy controls, access authentication, logging retention, secure updates, and user-consent mechanisms before deployment outside a research setting.

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APPENDICES

Appendix A Integrated interface specification

Contract	Type	Selected fields / semantics
MovementCommand	ROS 2 message	source, command, linear_speed, angular_speed, validity, emergency flag
ObstacleStatus	ROS 2 message	directional range/state observations and timestamp
SafetyState	ROS 2 message	NORMAL, CAUTION, RESTRICTED, EMERGENCY
RecognizedPerson	ROS 2 message	identity/recognition information for cross-component use
EmotionObservation	ROS 2 message	expression category/confidence context
SecurityEvent	ROS 2 message	event identity/type/status for security workflows
EmergencyStop / ResetEmergencyStop	ROS 2 services	explicit safety override and recovery
FollowPerson / ApproachCaller / PatrolRoute	ROS 2 actions	long-running autonomous behaviour contracts
Flask visual API	HTTP/JSON + MJPEG	unknown alerts, enrollment, emotion/identity status, camera/media and settings

Table A.1 Integrated interface summary

Appendix B Team contribution declaration

The team agrees that the three individual research components are distinct but interdependent. Ashandth Uthayashankar owns the offline voice intelligence research stream and contributed substantial system/safety/hardware integration. Sobiya Anton Suresh owns the visual-intelligence research stream and primarily implemented the Flutter companion application, with integration support from Ashandth. S. S. Y. Wickramasinghe owns the navigation, obstacle-avoidance, human-following, and path-planning research stream and contributed navigation/security requirements for the companion application together with the project website and dissemination material. Shared outputs include the final integrated robot, research paper, common architecture, testing activities, and final presentation preparation.

Member	Student ID	Primary individual contribution	Shared/integration contribution
Ashandth Uthayashankar	IT22272522	Offline voice intelligence, personalized interaction, safety-aware system integration	ROS 2 integration, motors/sensors/power, automatic startup, physical integration troubleshooting
Sobiya Anton Suresh	IT22203694	Visual intelligence, face/expression/unknown/object/fall functions, Flutter app	Camera-head integration, app-to-system integration, visual test evidence
S. S. Y. Wickramasinghe	IT21816086	Navigation, obstacle avoidance, human following, A* / 2D planning research	Navigation integration, app navigation/security requirements, project website/dissemination

Table B.1 Agreed individual and shared contribution mapping

Appendix C Integrated validation checklist

Capability	Evidence level at report preparation	Final-presentation update recommended
Voice wake + movement	Physical	Optional repeat/video evidence
Dynamic obstacle stop/resume	Physical	Repeat as integrated demo
Face following	Physical	Demonstrate alongside other running services if stable
Known/unknown + expression	Subsystem	Capture final robot-camera evidence
Unknown alert/enrollment	Subsystem app	Connect to final Pi service endpoint
Adaptive object learning	Subsystem	Repeat with one taught object on final camera
Fall detection	Implemented	Complete full-body validation before final presentation
A* / follow-me	Simulation	If time permits, connect to physical motion request path
Autonomous patrol/localisation	Continuing	Do not claim as complete without test evidence

Table C.1 Integrated validation checklist and remaining evidence

Appendix D Academic integrity and AI use disclosure

Generative AI and coding-assistance tools were used by the project team as development and writing support. ChatGPT was used for debugging explanations, architecture/diagram planning, document structuring, language refinement, synthesis of project evidence, and preparation of thesis/report drafts. GitHub Copilot was used by Sobiya Anton Suresh during parts of application/software development. AI-generated text or code suggestions were not treated as experimental evidence by themselves.

Technical claims in this common report were checked against the BUDDY research paper, the PP2 presentation, the current source repositories, supplied screenshots, physical test records, and the three individual thesis drafts. Unsupported quantitative claims were intentionally excluded. The team remains responsible for verifying the final report, references, code, data, diagrams, and contribution declarations before submission.

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